10963 - A Glimpse into Circumbinary Planet Formation at sub-100 au scales with HD 143811 AB b

Cycle: 5, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
Dr. Aneesh Baburaj (PI)	**Northwestern University**
Dr. Jason J. Wang (CoI)	Northwestern University
Dr. Jean-Baptiste Ruffio (CoI)	University of California - San Diego
Prof. Quinn Konopacky (CoI)	University of California - San Diego
Nathalie Jones (CoI)	Northwestern University
Rodrigo Ferrer-Chavez (CoI)	Northwestern University
Amanda Mia Chavez (CoI)	Northwestern University
Mr. Jayke Samson Nguyen (CoI)	University of California - San Diego
Dr. Kielan K. W. Hoch (CoI)	Space Telescope Science Institute
Dr. Chih-Chun Hsu (CoI)	Northwestern University
Dr. Alexander Bogdan Madurowicz (CoI)	Space Telescope Science Institute
William Balmer (CoI)	Space Telescope Science Institute
Dr. Genaro Suarez (CoI)	American Museum of Natural History
Dr. Robert J De Rosa (CoI) (ESA Member)	European Southern Observatory - Chile
Dr. Eric Nielsen (CoI)	New Mexico State University
Anne Elisabeth Peck (CoI)	New Mexico State University
Dr. Vito Squicciarini (CoI) (ESA Member)	University of Exeter
M. Christian Wilkinson (CoI) (ESA Member)	Observatoire de Paris
Dr. Anne-Marie Lagrange (CoI) (ESA Member)	Observatoire de Paris - Section de Meudon
Dr. Johan Mazoyer (CoI) (ESA Member)	LESIA, CNRS, Observatoire de Paris
Ms. Alice Radcliffe (CoI) (ESA Member)	Observatoire de Paris

Name	***Institution***
Dr. Mathilde Malin (CoI)	The Johns Hopkins University
Dr. Jerry Xuan (CoI)	University of California - Los Angeles
Dr. Yapeng Zhang (CoI)	California Institute of Technology

OBSERVATIONS

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
Observation Folder				
	1	HD-143811AB-b - Roll 1	NIRSpec IFU Spectroscopy	(2) HD-143811
	2	HD-143811AB-b - Roll 2	NIRSpec IFU Spectroscopy	(2) HD-143811

ABSTRACT

We propose a moderate-resolution spectroscopic study of the newly discovered directly imaged circumbinary planet HD 143811 AB b (M~6 MJup) using the JWST/NIRSpec IFU. At a projected separation of 60 au around a mid-F and late-F central binary pair, it is the only known wide orbit planet (Mp<10 MJup) orbiting a binary at <100 au. Recent work on planet formation in circumbinary disks indicates gravitational instability (GI) is more efficient in these disks, validated by stellar metal abundances for circumbinary companions like HIP 79098 AB b (sep~350 au). However, the <100 au separation of HD 143811 AB b implies core/pebble accretion cannot be discounted. Utilizing the G395H/F290LP mode mode of the NIRSpec IFU, we will obtain R2700 2.95.1 µm spectra, enabling studies of disequilibrium chemistry, measurement of molecular abundances corresponding to Carbon (C), Oxygen (O), and Sulfur (S)-bearing species and determination of metal enrichment. Stellar C,O,S abundances for HD 143811 AB b would provide the first test of formation indicators in a binary system that can be compared with current results from single star systems, allowing us to disentangle differences in planet formation between these environments. The comprehensive spectroscopic observations with JWST, not achievable with even the best ground-based facilities, would thus enable us to benchmark the next generation of planet formation models specifically aimed at explaining circumbinary planet formation.

OBSERVING DESCRIPTION

We propose to use the NIRSpec IFU to observe HD 143811 AB b using the G395H/F290LP mode to obtain moderate resolution (R2,700) spectra from 2.95.1 µm to measure abundances of H_2O, CO, CH_4, CO_2 and H_2S to precisions ~0.1 dex. The planet is 0.43" from the host star with a contrast ratio >1e-4, making it amenable to high-contrast observations with JWST NIRSpec.

No TA is required with the IFU. We split each visit in two observatory rolls (~10 deg apart) to avoid a chance alignment of the planet with bright speckles. We limit to 6 groups and use the NRSIRS2RAPID readout mode to prevent saturation while limiting detector noise. We choose 16 small cycling dithers and use 10 integrations to achieve the necessary S/N.

We provide a special Position Angle requirement (PA = 230245 deg, V3PA = 91-106 deg) to orient the star's diffraction spikes away from the planet. The PA chosen also satisfies the need to have the star on a different slice of the IFU to the planet, as the star is bright enough to partially saturate the slice of the NIRSpec IFU it falls on. We would like the two observations to be taken within 30 days of one another to reduce potential systematics while combining them.

Proposal 10963 - Targets - A Glimpse into Circumbinary Planet Formation at sub-100 au scales with HD 143811 AB b

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	HD-143811	RA: 16 03 33.4251 (240.8892713d) Dec: -30 08 13.38 (-30.13705d) Equinox: J2000	Proper Motion RA: -14.85 mas/yr Proper Motion Dec: -24.983999969663273 mas/yr Parallax: 0.0073065" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Detached binary stars, Exoplanet Systems, F stars]				
(3)	HD-143811AB-b	RA: 16 03 33.4381 (240.8893254d) Dec: -30 08 12.98 (-30.13694d) Equinox: J2000	Proper Motion RA: -14.85 mas/yr Proper Motion Dec: -24.983999969663273 mas/yr Parallax: 0.0073065" Epoch of Position: 2000	
Comments: Coordinates of the planet on 01/01/2027 are obtained from orbitize fits to current astrometry. We get separation ~ 431.41 mas and PA ~ 23.22 deg The companion coordinates are not used and are included just for visualization and cross-checking in Aladin. Category=Star Description=[Exoplanets]				

Observation

Proposal 10963, Observation 1: HD-143811AB-b - Roll 1 Fri Mar 13 20:04:06 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Data Excess over lower threshold

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(HD-143811AB-b - Roll 1 (Obs 1)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	HD-143811	RA: 16 03 33.4251 (240.8892713d) Dec: -30 08 13.38 (-30.13705d) Equinox: J2000	Proper Motion RA: -14.85 mas/yr Proper Motion Dec: -24.983999969663273 mas/yr Parallax: 0.0073065" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Detached binary stars, Exoplanet Systems, F stars]

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	CYCLING	SMALL	1	16	

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395H/F290LP	NRSIRS2RAPID	6	10	false	true	NONE	16	160	16339.557	268778

Special Requirements

Aperture PA Range 229.97164917 to 244.97164917 Degrees (V3 91.0 to 106.0)

Group Observations 1, 2 within 30 Days
Aperture PA Offset 2 from 1 by 10 to 14 Degrees (Same offsets in V3)

Observation

Proposal 10963, Observation 2: HD-143811AB-b - Roll 2

Fri Mar 13 20:04:06 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 2:1) Warning (Form): Data Excess over lower threshold

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(HD-143811AB-b - Roll 2 (Obs 2)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	HD-143811	RA: 16 03 33.4251 (240.8892713d) Dec: -30 08 13.38 (-30.13705d) Equinox: J2000	Proper Motion RA: -14.85 mas/yr Proper Motion Dec: -24.983999969663273 mas/yr Parallax: 0.0073065" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Detached binary stars, Exoplanet Systems, F stars]

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	CYCLING	SMALL	1	16	

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395H/F290LP	NRSIRS2RAPID	6	10	false	true	NONE	16	160	16339.557	268778

Special Requirements

Group Observations 1, 2 within 30 Days
Aperture PA Offset 2 from 1 by 10 to 14 Degrees (Same offsets in V3)